Computer Science Course — ENS Year 1 — Session 02 (EN)

Instructor: BELACEL Madani

Institution: Higher Normal School of Mostaganem

Module: Computer Science

Level: ENS — Year 1 — English Department

Duration: 1h30

Academic year: 2025-2026

ICT Course 02 — Computer Science and Computer Architecture

Teacher: BELACEL Madani

Module: ICT

Level: License 1 — French Department

Duration: 1h30

Academic year: 2025-2026

(Email and detailed progress of the session: see PDF cover page.)

Course Plan

1. Introduction: Why understand computers?
2. Definition of IT
3. The computer: concept and model of von Neumann
4. Hardware components
5. Software and operating systems (Software)
6. Interaction between Hardware and Software
7. Best practices for university students
8. Conclusion and transition to Windows

1. Introduction *(10 min)*

Understanding how the computer works is essential for any student, even in the Languages sector. This allows you to better use your tools (Word, browser, Moodle), avoid data loss, diagnose simple problems and effectively organize your digital workspace.

The computer has become **the main university working instrument**.

2. What is IT? *(10 mins)*

Computer science is the science of **automatic information processing** using algorithms executed by machines. It includes theory, hardware, software and networks.

User approach: understand enough to use the computer well in your studies.

3. von Neumann model *(15 min)*

A computer is a programmable machine capable of: receiving data (**Input**), processing it (**CPU**), temporarily storing it (**RAM**), returning it (**Output**) and retaining it (**Secondary storage**).

This operation follows the **von Neumann model** (1945), still used today.

4. Hardware Components *(15 min)*

• **Processor (CPU)**: the brain of the computer

• **Motherboard**: interconnection of components

• **RAM**: volatile RAM, very fast

• **Storage**: HDD or SSD (faster)

• **Peripherals**: input (keyboard, mouse), output (screen, printer), external storage (USB key)

Important point: RAM loses unsaved data when turned off.

5. Software *(15 min)*

• **Operating system**: Windows, Linux, macOS

• **Software application**: Word, PowerPoint, browsers, Moodle

• **Utility software**: antivirus, compressors

The hardware is tangible; the software is intangible but essential.

6. Hardware / Software Interaction *(10 min)*

Example: keyboard keystroke → OS translated → processor processes → screen displays.

Without software, hardware is useless. Without hardware, the software cannot run.

7. Good Practices *(10 min)*

1. Backup regularly (`Ctrl + S` + Cloud)
2. Organize your files from the start of the semester
3. Update Windows and software
4. Use an antivirus
5. Distinguish between RAM and permanent storage

8. Conclusion *(5 min)*

Understanding computer architecture allows you to be more independent and efficient.

Next lesson: Windows — Interface, files and customization.